

Statistical Analysis of Biomarkers for Autoimmune Disease Diagnosis

Aishwarya Mohanraj · Aiswarya Perumbilly · Aishwarya Thudukuchi Srimurugan · Damaris Adeniji

Luddy School of Informatics, Computing & Engineering · Indiana University · Group 1

N = 12,500 Patients

AUC = 0.999

CRP · ESR · Hemoglobin

Study Overview

Research Question:

How do inflammatory biomarkers and demographic factors differ between individuals with and without autoimmune disease, and can these variables accurately predict disease status?



Dataset

Kaggle: Comprehensive Autoimmune Disorder Dataset (Ragheb, 2024)
12,500 patients total
10,000 Autoimmune | 2,500 Healthy
8 laboratory biomarkers
Synthetic clinical distributions



Objective

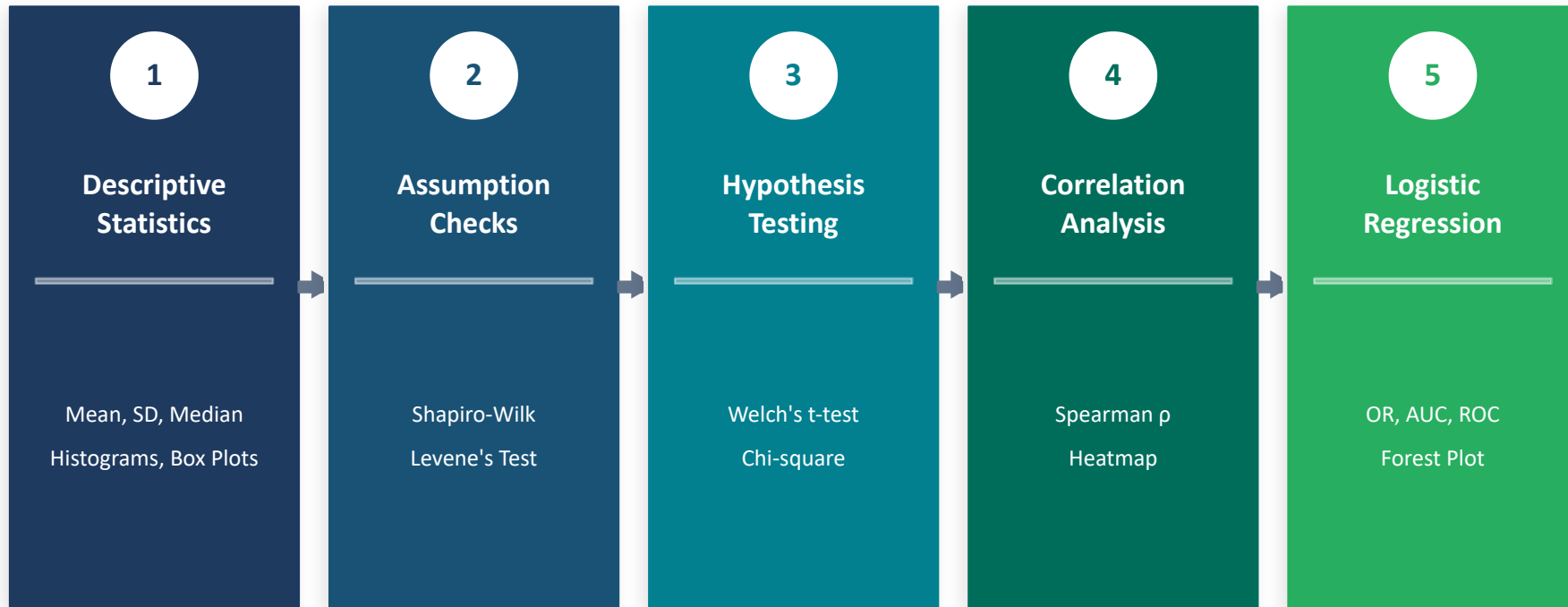
Distinguish autoimmune vs. healthy patients using routine blood biomarkers
Identify strongest predictors
Build a predictive model
Assess demographic effects (age, gender)



Approach

Descriptive Statistics
Welch's t-tests
Chi-square & Proportion Tests
Spearman Correlation
Binary Logistic Regression

Analytical Pipeline



All analyses conducted in R (tidyverse, broom, pROC, kableExtra) · Significance level: $\alpha = 0.05$

Dataset & Descriptive Statistics

Dataset Composition

12,500

Total Patients

10,000 (80%)

Autoimmune

2,500 (20%)

Healthy Controls

49.8 years

Mean Age

Gender Balance

♂ Male 50.5% (n=6,316)

♀ Female 49.5% (n=6,184)

Key Biomarker Means by Group

Biomarker	Autoimmune	Healthy
ESR (mm/hr)	29.24	9.26
CRP (mg/L)	5.44	0.49
Hemoglobin (g/dL)	12.99	13.58
WBC (/μL)	7,956	7,777
Neutrophils (%)	53.22	55.58
Lymphocytes (%)	30.11	30.07
C3 (g/L)	0.95	1.00
C4 (g/L)	0.35	0.30

Red = elevated vs healthy · Green = lower risk direction

Hypothesis Testing — Welch's t-Tests

H_0 : No difference in biomarker means between Autoimmune and Healthy groups · $\alpha = 0.05$ · Welch's t-test used due to unequal variances (Levene's $p < 0.05$)

CRP (mg/L)		H: 0.49 AI: 5.44	$t = -183.45$ $\checkmark p < 0.001$
ESR (mm/hr)		H: 9.26 AI: 29.24	$t = -121.83$ $\checkmark p < 0.001$
Hemoglobin (g/dL)		H: 13.58 AI: 12.99	$t = 17.42$ $\checkmark p < 0.001$
Neutrophils (%)		H: 55.58 AI: 53.22	$t = 12.30$ $\checkmark p < 0.001$
C4 (g/L)		H: 0.3 AI: 0.35	$t = -24.02$ $\checkmark p < 0.001$
C3 (g/L)		H: 1 AI: 0.95	$t = 10.83$ $\checkmark p < 0.001$
WBC ($\mu\text{L} \div 1000$)		H: 7.777 AI: 7.956	$t = -4.06$ $\checkmark p < 0.001$
Lymphocytes (%)		H: 30.07 AI: 30.11	$t = 0.24$ $\times p = 0.812$

■ Healthy Controls
 ■ Autoimmune (elevated)
 ■ Non-significant

$\checkmark$ = Significant | $\times$ = Not Significant

Spearman Correlation Analysis

Why Spearman?

- Non-normal, right-skewed distributions (Shapiro-Wilk $p < 0.05$ for all)
- Robust to outliers common in inflammatory data
- Captures monotonic (not just linear) relationships

Key Correlations

$\rho = +0.40$ ESR $\leftrightarrow$ CRP

Both reflect systemic inflammation

$\rho = -0.41$ ESR $\leftrightarrow$ Hemoglobin

Inflammation suppresses erythropoiesis

$\rho = -0.43$ CRP $\leftrightarrow$ Hemoglobin

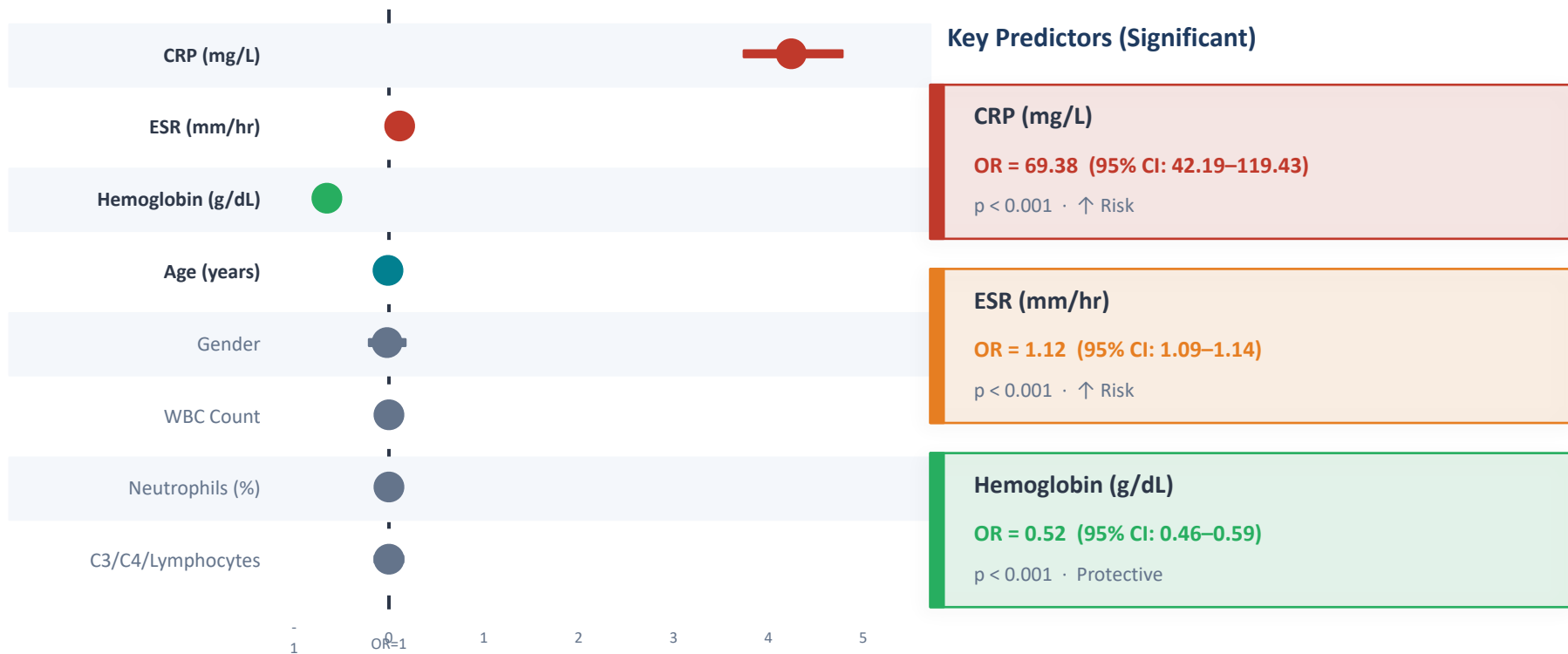
Anaemia of chronic inflammation

All remaining $|\rho| < 0.15$ — biomarkers largely independent $\rightarrow$ no multicollinearity concern

	ESR	CRP	Hgb	WBC	Neut	Lymph	C3	C4
ESR	1.00	0.40	-0.41	0.01	-0.05	0.01	-0.04	0.09
CRP	0.40	1.00	-0.43	0.02	-0.07	0.00	-0.04	0.10
Hgb	-0.41	-0.43	1.00	-0.02	0.06	0.00	0.03	-0.08
WBC	0.01	0.02	-0.02	1.00	-0.01	-0.01	0.00	0.01
Neut	-0.05	-0.07	0.06	-0.01	1.00	0.01	0.00	-0.02
Lymph	0.01	0.00	0.00	-0.01	0.01	1.00	0.00	0.00
C3	-0.04	-0.04	0.03	0.00	0.00	0.00	1.00	-0.01
C4	0.09	0.10	-0.08	0.01	-0.02	0.00	-0.01	1.00

Binary Logistic Regression — Predictors of Autoimmune Disease

Full Dataset $n = 12,500$ · Outcome: Autoimmune Diagnosis (binary) · Null deviance: 12,510 → Residual deviance: 1,145 · AIC: 1,167

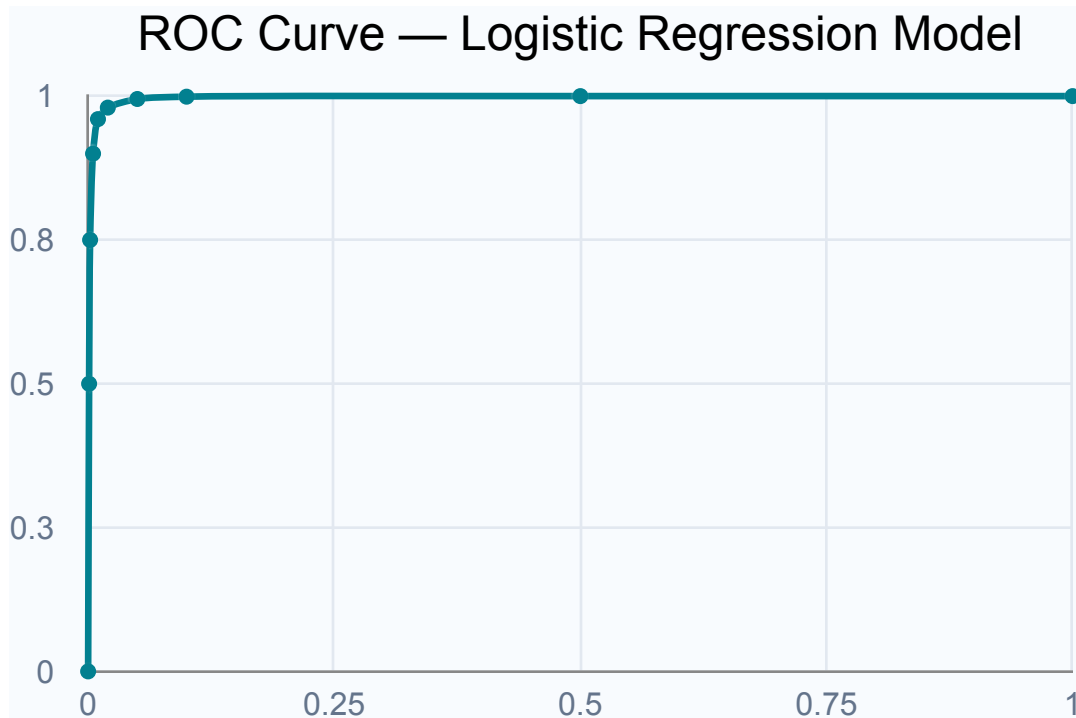


Model Performance & ROC Analysis



Interpretation

Driven by strongly separated CRP & ESR distributions in synthetic data. Real-world AUC typically ~0.80–0.90.



Discussion — Key Findings

CRP — Dominant Predictor

OR = 69.38. Acute-phase reactant produced by liver within hours of immune activation. Most sensitive proxy for systemic inflammation — the hallmark of autoimmune disease.

ESR — Independent Predictor

OR = 1.12 per unit increase. Captures aggregation of red blood cells driven by inflammatory plasma proteins. Provides additional independent predictive value beyond CRP.

Hemoglobin — Protective Marker

OR = 0.52. Each +1 g/dL → ~48% lower disease odds. Reflects anaemia of chronic inflammation: hepcidin elevation, disrupted iron homeostasis, suppressed erythropoiesis.

Lymphocytes — Non-Significant

t = 0.24, p = 0.812. Autoimmune dysregulation involves specific lymphocyte subtypes (autoreactive T-cells, regulatory T-cells, self-reactive B-cells) — not captured by total count.

Gender — No Significant Effect

χ^2 p = 0.511. Inconsistent with real-world epidemiology (e.g., Lupus 9:1 F:M; RA 3:1). Likely reflects synthetic data generation limitation — no gender bias was encoded.

Age — Trivial Effect

OR $\approx$ 0.99, $\Delta \approx$ 1.5 years between groups. Illustrates that with N > 12,000, even negligible differences reach statistical significance. Clinical relevance is negligible.

Limitations & Future Directions

Limitations

- 1 Synthetic data — does not replicate real-world clinical variability or established gender disparities
- 2 Cross-sectional design — identifies associations, not causal relationships
- 3 Near-perfect AUC likely artefact of synthetic data generation
- 4 All autoimmune conditions collapsed to a single binary outcome — disease-specific biomarker profiles differ



Future Directions



Real Patient Data

Validate using clinical biobanks or electronic health records



Autoantibody Markers

Add anti-dsDNA, rheumatoid factor, anti-CCP for biological specificity



Genetic Variables

HLA haplotype data + longitudinal follow-up for causal inference



Machine Learning

Random forests & gradient boosting to detect non-linear biomarker interactions

Conclusion

- CRP, ESR, and Hemoglobin are the most informative biomarkers — consistent across all statistical methods applied.
- CRP is the single strongest predictor (OR = 69.38), with ESR providing independent additional predictive value.
- Hemoglobin acts as a significant protective inverse marker (OR = 0.52) — anaemia of chronic inflammation.
- Demographic variables (age, gender) contribute minimally to disease discrimination in this dataset.
- AUC = 0.999 reflects near-perfect model performance attributable to synthetic data characteristics.
- These results support routine inflammatory marker testing for early autoimmune disease detection.